IMAGE SEGMENTATION
(Using K-means clustering)

1 AIM

To implement and evaluate K-means clustering-based image segmentation for precise and efficient partitioning of visual elements, enhancing image analysis and interpretation

2 METHODOLOGIES

2.1 K-means Clustering

K-means clustering is an unsupervised machine learning algorithm used to partition data points into a specified number distinct, non-overlapping clusters, denoted as 'k'. It groups data based on their inherent similarities, aiming to minimize the distance between data points within each cluster and their respective centroids.

This is suitable for image segmentation primarily due to its simplicity, efficiency, and effectiveness in grouping similar pixels into clusters based on their feature similarity, such as colour or intensity, effectively isolating different objects or regions within the image.

2.1.1 Algorithm

K-means clustering algorithm is an iterative method that partitions a dataset into k clusters by minimizing the sum of squared distances between data points and their assigned cluster centroids. The steps include:

1. <u>Choose the number of clusters (k)</u>

   Determine the number of clusters into which the dataset or image should be divided. The value of 'k' is usually chosen based on prior knowledge or by experimenting with different values.

2. <u>Initialise cluster centroids</u>

   Randomly select k data points (pixels, in the case of image) as the initial centroids representing each cluster. These centroids act as the starting points for the clustering process.

3. <u>Assign data points to the nearest centroid</u>

   For every data point, calculate the distance to each centroid (commonly used Euclidean distance) and assign the point to the cluster with the closest centroid. This forms K initial clusters based on colour similarity.

4. Update the centroids

   After all data points are assigned, compute the new centroid of each cluster by taking the mean of all points belonging to that cluster. These updated centroids represent the current clusters.

5. Repeat steps 3 and 4

   Repeat the assignment and centroid update steps until the centroids no longer change significantly between iterations or until a maximum number of iterations is reached. At this stage, the algorithm is said to have converged, and the final clusters represent the segmented regions.

2.2 Image Segmentation using K-means Clustering[4]

Image segmentation is the process of dividing an image into distinct regions that share similar characteristics such as colour, intensity or texture. In this project, image segmentation is achieved using the K-means clustering algorithm explained above. It is suitable for image segmentation primarily due to its simplicity, efficiency, and effectiveness in grouping similar pixels into clusters based on their feature similarity, such as colour or intensity, effectively isolating different objects or regions within the image.

2.3 About dataset[1]

The dataset was found from a website called Kaggle and the link for the same is provided in the references section. It consists of 500 natural images, ground-truth human annotations and benchmarking code. The data is explicitly separated into disjoint train, test and validation subsets. The dataset is an extension of the BSDS300, where the original 300 images are used for training / validation and 200 fresh images, together with human annotations, are added for testing.

Since K-means clustering is an unsupervised machine learning algorithm, the training and testing of the model is not required. Hence, only the training data (200 images) was used.

2.4 LAB Colour Space

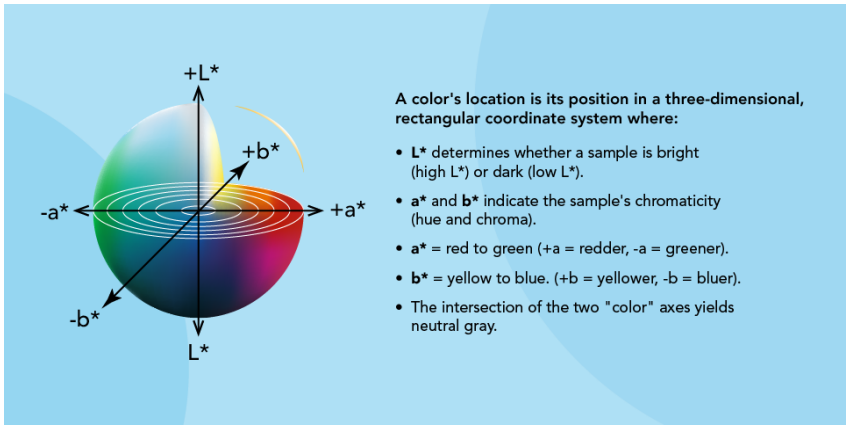
LAB colour space is a way to represent colours that is closer to how humans see them, making it easier to distinguish differences in colour.

It has three parts:

- ‘L’ for lightness (from black at 0 to white at 100)
- ‘A’ represents colours from green (negative values) to red (positive values)
- ‘B’ represents colours from blue (negative values) to yellow (positive values)

Unlike RGB, which mixes red, green, and blue light directly, LAB separates lightness from colour information and arranges colours so the changes in values correspond

better to what people perceive. This makes LAB colour space very useful in performing tasks such as image segmentation.



Many image processing libraries, like OpenCV, use BGR (RGB and BGR represent the same colours but store the pixel channels in reverse order) as the default ordering because of how some cameras and image formats store pixel data.

When an image is segmented (using K-means clustering) in the RGB (or BGR) format, pixel colours get grouped based on their distance in this space, which mixes lightness (brightness) and colour information together. But for real-world photographs, images with gradual shading, uneven lighting, or complex colour distributions, the segmentation may not align well with human perception or clear object boundaries. Hence, we convert BGR format (the default format used by OpenCV) to LAB Colour Space.

2.5 Morphological Operations[3]

Morphological operations are special image-processing techniques that focus on the shape and structure of objects within an image, instead of just their colour or brightness. They are mainly used to clean up, modify, or analyse binary (black-and-white) images, but can also be applied to grayscale images.

Morphological operations use a small pattern called a structuring element (like a 3x3 or 5x5 square or circle) to examine and modify pixels based on their arrangement in the image.

The main operations are:

- **Erosion:** Shrinks objects by removing pixels on boundaries, helpful for getting rid of small white noise or separating joined shapes.
- **Dilation:** Expands objects by adding pixels to boundaries, helpful for filling in small holes or connecting disjoint objects.

- **Opening:** Erosion followed by dilation; cleans up small objects or noise while keeping larger shapes intact.
- **Closing:** Dilation followed by erosion; fills small holes or gaps within objects.

These operations are widely used in image segmentation, object detection, and noise reduction. They help create cleaner regions, smoother boundaries, and more meaningful shapes

Morphological operations generally show more visible effects on binary or grayscale images because they either remove noise, smooth boundaries, expand or shrink bright/dark regions based on the structuring element applied.

2.6 Applications of Image Segmentation using K-means clustering

- **Medical Imaging:** Used to separate anatomical structures like tumours, tissues, organs etc. in MRI, CT and X-ray images.
- **Image Compression:** Reduces the number of colours in an image by clustering similar colour pixels, minimizing storage size without significant quality loss.
- **Object Detection and Recognition:** Helps in identifying and isolating objects of interest from the background in computer vision tasks.
- **Background removal and Image editing:** Enables automatic separation of foreground objects from the background for photo editing and compositing.

3 CODE

4 RESULTS

Figure 1 shows the first original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). The original image consists of at least more than 5 colours but since the value 'k' is 3, only 3 clusters of different colours are formed in the segmented image.

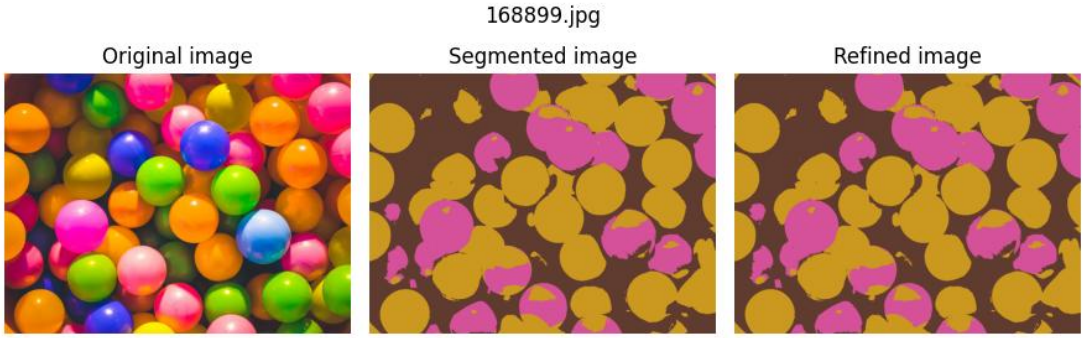


Figure 1: Comparison of Original, K-means Segmented and Morphologically Refined Image 1

Figure 2 shows the second original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). There are 3 major colours seen in the original image i.e., red, yellow and green. But when the image is segmented or when clusters are formed, apart from the major colours, there's a separate cluster formed for the shadows.

Figure 2: Comparison of Original, K-means Segmented and Morphologically Refined Image 2

Figure 3 shows the third original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). The original image consists of a lot of colours. When segmented, a lot of colours that were close to each other were clustered into one as only 3 clusters were to be formed.

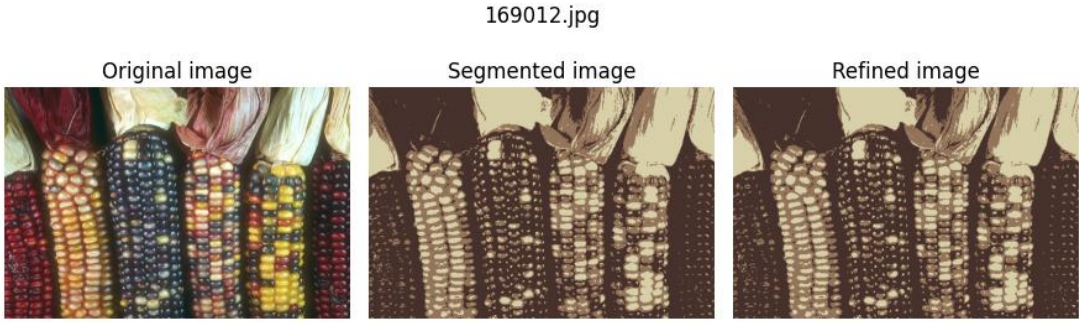


Figure 3: Comparison of Original, K-means Segmented
and Morphologically Refined Image 3

Figure 4 shows the fourth original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). The original image consists of major components such as the sky, the grass, the house and the lady. The colour of the sky is similar to that of the lady's dress and hence it was segmented into one cluster.

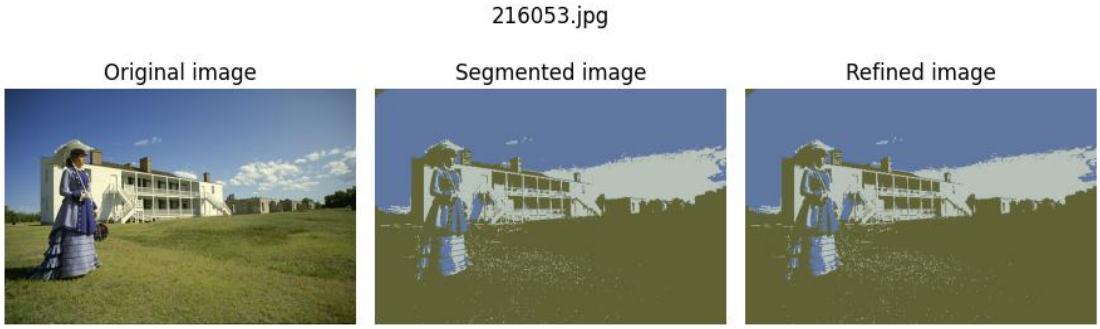


Figure 4: Comparison of Original, K-means Segmented
and Morphologically Refined Image 4

Figure 5 shows the fifth original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). There are 2 major colours seen in the original image i.e., green and brown, and a little bit of yellow. Hence, this image was easily segmented into 3 clusters.

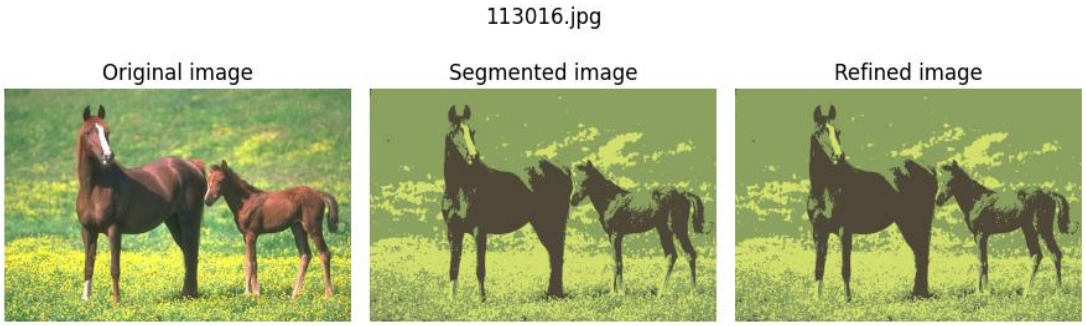


Figure 5: Comparison of Original, K-means Segmented
and Morphologically Refined Image 5

Figure 6 shows the sixth original image, the segmented image using K-means clustering (K = 3) and finally the refined image which was obtained by performing morphological operations (here, dilation). The major colours seen in the image are light green, dark green, light blue (sky) and dark blue (water). When segmented, the dark green and dark blue were segmented into the same cluster.

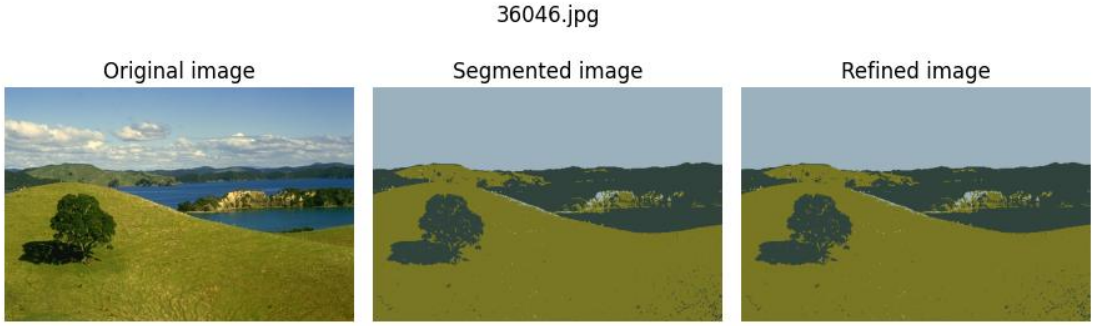


Figure 6: Comparison of Original, K-means Segmented and Morphologically Refined Image 6

5 DISCUSSION

The results clearly demonstrate the effectiveness of the K-means clustering algorithm in performing image segmentation based on colour similarity. For all six images, K-means successfully grouped pixels with similar colour intensities into three distinct clusters, as the value of K is 3.

In the case of images with a wide range of colours i.e., Figures 1 and 3, multiple similar shades were merged into single clusters because the algorithm was constrained to form only three colour groups. This led to a slight loss of precise colour variation but enhanced the clarity of dominant regions.

For images with fewer dominant colours such as Figures 2 and 5, segmentation results were more accurate and visually closer to the original image, as each major colour was assigned to a separate cluster.

For natural scenes i.e., Figures 4 and 6, K-means was able to distinguish between large regions such as the sky, grass, and objects. However, due to similarity in hue or brightness, certain regions (e.g., the sky and the dress in Figure 4 or the dark water and vegetation in Figure 6) were grouped into the same cluster. This behaviour highlights one of the known limitations of K-means, it relies on colour distance alone, without spatial or texture awareness.

In this project, morphological dilation was chosen as the post-processing technique since the objective was to clean the boundaries of the segmented regions. Dilation expands the segmented areas slightly, thereby filling small edge gaps and creating more continuous object outlines. Other operations such as erosion or opening were not used, as they tend to shrink or break boundaries, which contradicts the goal of boundary cleaning. The use of dilation thus preserves object integrity while improving the visual smoothness of the segmentation output.

6 CONCLUSION

In this assignment, image segmentation was successfully performed using K-means clustering algorithm in the Lab colour space, followed by morphological post-processing. The algorithm effectively grouped pixels with similar colour intensities into distinct regions, thereby simplifying the image and highlighting its major components.

Morphological dilation was applied as a post-processing step to clean and enhance the boundaries of the segmented regions. Since the K-means output was already clean and continuous, the refinement produced only subtle visual differences. Morphological operations are designed to work on grayscale images. It is based on pixel intensity values and a structuring element. They look at how pixel brightness (gray levels) changes in a neighbourhood. If morphological operations are directly applied to an RGB image, the operation is constrained to handle three different intensities at once, there's no clear definition of minimum or maximum pixel because each channel might behave differently and hence it can lead to colour distortion or unpredictable results, since operations on each colour plane independently might alter hues. This is why, before performing morphological operations, the RGB image is converted to a binary image.

Overall, the combination of K-means clustering and morphological dilation provides a simple, efficient, and effective approach for colour-based image segmentation. Future improvements could involve incorporating spatial features or adaptive morphological operations to handle complex textures or overlapping colour regions more effectively.

7 REFERENCES

[1] **Dataset:** Kaggle, "Berkeley Segmentation Dataset 500 (BSDS500)," Kaggle, 2005. [Online].
Available: https://www.kaggle.com/datasets/balraj98/berkeley-segmentation-dataset-500-bsds500?resource=download.

[2] OpenCV Team, "OpenCV-Python Tutorials," OpenCV Documentation, 2025. Available: https://docs.opencv.org/4.x/d6/d00/tutorial_py_root.html

[3] R. Kimmel and J. A. B. Tenenbaum, "Efficient Dilation, Erosion, Opening and Closing Algorithms," IEEE Trans. Pattern Anal. Mach. Intell., 2002.

[4] J. Mukhopadhyay et al., "Image segmentation based on K-mean algorithm," J. Inf. Hiding Multimedia Signal Process., 2018.